

FTTH Network Planning, GIS Infrastructure Analytics and GPON Engineering Design

Phase 6 DHA – Fiber To The Home (FTTH) Network Deployment Project

Technologies Used

- QGIS
- ArcMap
- Power BI
- Microsoft Excel
- GPON Network Planning
- GIS Spatial Analysis
- FTTH Design and Engineering
- Telecom Infrastructure Analytics

Executive Summary

The telecommunications industry is experiencing unprecedented growth due to increasing demand for high-speed internet, cloud services, video streaming, smart devices, and digital transformation initiatives. To meet these requirements, network operators worldwide are investing heavily in Fiber to the Home (FTTH) infrastructure, which provides significantly higher bandwidth, lower latency, and greater scalability than traditional copper-based access networks.

This project presents a complete FTTH network planning and GIS infrastructure analytics workflow developed for a residential and commercial service area. The project integrates telecom engineering principles, Geographic Information Systems (GIS), optical network design, serviceability analysis, infrastructure planning, and business intelligence reporting into a unified engineering framework.

The network design follows modern Gigabit Passive Optical Network (GPON) architecture and includes feeder network planning, distribution network design, Fiber Distribution Hub (FDH) allocation, Fiber Access Terminal (FAT) planning, Network Access Point (NAP) deployment, optical loss budgeting, duct infrastructure analysis, and service coverage assessment.

To support engineering decision-making and project management activities, Power BI dashboards were developed to visualize network infrastructure metrics, capacity utilization, serviceability indicators, and operational Key Performance Indicators (KPIs).

This project demonstrates how GIS, telecom engineering, and data analytics can be integrated to support the planning, deployment, monitoring, and future expansion of modern fiber optic networks.

About the Author

I am currently pursuing a Master of Science in Geoinformatics & Data Science at the University of Milan, Italy, while building upon several years of professional experience in GIS, telecom engineering, infrastructure planning, and spatial analytics.

My academic background combines scientific research, geospatial technologies, and data-driven decision-making. In addition to my graduate studies, I have completed advanced training in Data Science and Artificial Intelligence, which has strengthened my ability to analyze complex datasets and develop analytical solutions for real-world engineering problems.

Professionally, I have worked as a GIS Analyst, GIS Surveyor, and FTTH Network & Data Specialist on projects involving telecommunications infrastructure, utility mapping, spatial database development, fiber network planning, and business intelligence reporting. My work has included High-Level Design (HLD), Low-Level Design (LLD), GIS data management, FTTH planning, Power BI dashboards, Spatial Analysis and telecom infrastructure analytics.

The purpose of this project is to demonstrate the practical integration of GIS technologies and telecom engineering principles within a complete FTTH planning workflow while displaying the analytical and technical capabilities required for modern broadband infrastructure development.

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1. Introduction

Access to reliable broadband connectivity has become a fundamental requirement for economic growth, education, healthcare, commerce, and communication. As digital technologies continue to evolve, traditional copper-based communication networks face increasing limitations in terms of bandwidth capacity, transmission distance, reliability, and maintenance requirements.

Fiber optic technology has emerged as the preferred solution for next-generation broadband infrastructure because of its ability to transmit large volumes of data over long distances with minimal signal degradation. Compared to copper networks, optical fiber offers significantly greater bandwidth capacity, lower attenuation, immunity to electromagnetic interference, improved reliability, and a longer operational lifespan.

Among the various fiber deployment architectures, Fiber to the Home (FTTH) provides the highest quality of service by extending fiber connectivity directly to customer premises. This architecture enables service providers to deliver high-speed internet, IPTV, voice services, cloud applications, and future digital services through a single optical access network.

Most modern FTTH deployments are based on Gigabit Passive Optical Network (GPON) technology. GPON allows multiple customers to share the same fiber infrastructure through passive optical splitters, reducing deployment costs while maintaining high network performance.

The objective of this project was to develop a complete FTTH planning framework using GIS-based methodologies and telecom engineering principles. The project focuses on designing a scalable and serviceable fiber network capable of supporting current demand while providing sufficient capacity for future expansion.

2. Project Objectives

The primary objectives of this project were:

- Design a scalable FTTH network architecture.
 - Develop feeder and distribution fiber infrastructure.
 - Create High-Level Design (HLD) and Low-Level Design (LLD) maps.
 - Define FDH, FAT, and NAP service boundaries.
 - Analyze serviceability and customer coverage.
 - Estimate infrastructure requirements through BOQ development.
 - Evaluate fiber capacity and utilization.
 - Develop engineering dashboards for operational reporting.
 - Integrate GIS and telecom engineering workflows into a single planning framework.
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3. Fundamentals of Fiber Optic Communication

Fiber optic communication is based on the transmission of light signals through thin strands of glass known as optical fibers. Information is encoded into light pulses and transmitted through the fiber core using the principle of total internal reflection.

A typical optical fiber consists of three major components:

Core

The central portion of the fiber through which light travels.

Cladding

A surrounding layer with a lower refractive index that keeps light confined within the core.

Protective Coating

An external protective layer that shields the fiber from physical damage and environmental conditions.

The use of optical signals instead of electrical signals enables significantly higher transmission speeds and lower signal losses over long distances.

Major advantages of fiber optics include:

- Extremely high bandwidth capacity.
- Low signal attenuation.
- Resistance to electromagnetic interference.
- Enhanced reliability.
- Long operational lifespan.
- Reduced maintenance requirements.

These characteristics make fiber optics the preferred medium for modern broadband access networks.

4. GPON Architecture

Gigabit Passive Optical Network (GPON) is a Point-to-Multipoint (P2MP) access technology widely used in FTTH deployments.

A GPON network consists of three primary components:

Optical Line Terminal (OLT)

The OLT is installed at the central office or exchange and serves as the main transmission equipment. It manages traffic, controls subscriber connections, and distributes services across the optical network.

Optical Distribution Network (ODN)

The ODN represents the passive portion of the network and includes:

- Feeder Fibers
- Distribution Fibers
- Splitters

- FDHs
- FATs
- NAPs
- Splice Closures

The ODN is responsible for transporting optical signals from the OLT to customer locations.

Optical Network Terminal (ONT)

The ONT is installed at the customer premises and converts optical signals into usable voice, data, and video services.

The GPON architecture allows multiple subscribers to share a single fiber infrastructure while maintaining efficient utilization of network resources and reducing deployment costs.

5. FTTH Network Hierarchy

A well-designed FTTH network follows a hierarchical structure to ensure scalability, maintainability, and operational efficiency.

Feeder Network

The feeder network serves as the backbone of the FTTH infrastructure. It connects the OLT to Fiber Distribution Hubs (FDHs) and carries aggregated traffic for multiple service areas.

Typical feeder cable capacities include:

- 48 Fiber
- 96 Fiber
- 144 Fiber

The feeder network is designed with spare capacity to support future growth and network expansion.

Distribution Network

The distribution network extends connectivity from FDHs to FATs and NAPs located throughout the service area.

Typical distribution cable capacities include:

- 12 Fiber
- 24 Fiber
- 48 Fiber

The distribution network provides localized service coverage and customer access.

Drop Network

The drop network represents the final connection between the NAP and the customer premises. It typically utilizes low-count drop cables and is designed to provide reliable service connectivity while minimizing installation complexity.

6. Field Survey and Data Collection

Accurate field data collection is one of the most critical stages of FTTH network planning. The quality of network design depends heavily on the accuracy of the information collected during site surveys.

Before developing the High-Level Design (HLD) and Low-Level Design (LLD), a detailed field survey was conducted to understand the existing infrastructure, service demand, accessibility constraints, and deployment feasibility within the study area.

The survey activities focused on identifying existing telecom assets, underground utility routes, customer concentrations, road networks, right-of-way conditions, and potential locations for fiber infrastructure deployment.

Survey Objectives

The primary objectives of the survey were:

- Identify existing telecom infrastructure.
- Verify road and street accessibility.
- Determine suitable FDH locations.
- Identify FAT and NAP service zones.
- Assess duct and conduit availability.
- Evaluate customer density and service demand.
- Verify utility crossings and physical obstacles.
- Support BOQ estimation and deployment planning.

Existing Infrastructure Assessment

Existing infrastructure was reviewed to determine opportunities for infrastructure reuse and cost optimization.

The survey included identification of:

- Existing fiber routes.
- Joint boxes and handholes.
- Utility poles.
- Duct networks.
- Existing telecom cabinets.
- Splice closures.
- Existing customer access infrastructure.

Where feasible, existing infrastructure was incorporated into the design to reduce civil works and deployment costs.

GPS and Spatial Data Collection

Geographic coordinates of critical infrastructure elements were collected and validated for use within the GIS environment.

Spatial datasets were organized into thematic layers including:

- Feeder Network
- Distribution Network
- Duct Infrastructure
- FDH Locations
- FAT Locations
- NAP Locations
- Service Boundaries

The collected information was integrated into QGIS and ArcGIS environments for further analysis and engineering design.

Demand Assessment

Customer demand plays a significant role in FTTH network planning.

The survey evaluated:

- Residential developments.
- Commercial zones.
- Multi-story buildings.
- Educational institutions.
- Government facilities.
- Business centers.
- Future development areas.

This information was used to estimate Homes Passed, future subscriber growth, and infrastructure capacity requirements.

Engineering Constraints

Several engineering constraints were identified during the survey process including:

- Limited duct availability.
- Congested utility corridors.
- Road crossings.
- High-density urban development.
- Utility conflicts.
- Access limitations.

These constraints were incorporated into the final design to ensure constructability and operational efficiency.

Survey Outcomes

The field survey provided the foundation for all subsequent planning activities including:

- HLD Development

- LLD Development
- Serviceability Analysis
- BOQ Preparation
- Dashboard Analytics
- Capacity Planning

By combining field observations with GIS analysis, the project achieved a more accurate and practical representation of the real-world network environment.

7. Optical Power Budget and Link Loss Analysis

One of the most critical aspects of FTTH network planning is ensuring that sufficient optical power reaches every customer connected to the network. Unlike traditional electrical communication systems, optical signals gradually lose strength as they travel through fiber cables and network components. This reduction in signal strength is known as attenuation or optical loss.

If the total optical loss exceeds the allowable limits of the GPON system, customers may experience poor service quality, intermittent connectivity, reduced bandwidth, or complete service outages. Therefore, optical power budgeting is an essential step in network design.

The total optical loss in a GPON network is influenced by several components including fiber attenuation, splice losses, connector losses, and splitter losses.

Fiber Attenuation

Optical fiber naturally attenuates light signals as they travel through the cable. Typical attenuation values for single-mode fiber are:

- 1310 nm wavelength: approximately 0.35 dB/km
- 1490 nm wavelength: approximately 0.25 dB/km
- 1550 nm wavelength: approximately 0.20 dB/km

The longer the fiber route, the greater the cumulative loss.

Splice Loss

Fusion splicing is commonly used to join fiber segments together. Although modern fusion splicing techniques are highly efficient, every splice introduces a small amount of signal loss.

Typical splice loss:

- 0.05 dB to 0.10 dB per splice

Connector Loss

Connectors are used wherever fibers need to be disconnected or terminated. Each connector introduces additional attenuation.

Typical connector loss:

- 0.20 dB to 0.50 dB per connector

Splitter Loss

Passive optical splitters divide optical signals among multiple customers. While splitters eliminate the need for active equipment in the field, they contribute the largest portion of optical loss within GPON networks.

Typical splitter losses include:

- 1:2 Splitter $\approx$ 3.5 dB
- 1:4 Splitter $\approx$ 7.2 dB
- 1:8 Splitter $\approx$ 10.5 dB
- 1:16 Splitter $\approx$ 13.5 dB
- 1:32 Splitter $\approx$ 17.2 dB
- 1:64 Splitter $\approx$ 21 dB

The selection of splitter ratios directly affects network reach, customer density, and service quality.

Link Loss Budget Methodology

The optical power budget for each service area was evaluated using the following approach:

Total Loss = Fiber Loss + Splitter Loss + Connector Loss + Splice Loss

The objective was to ensure that total losses remain within acceptable GPON operating limits while maintaining adequate design margins for future maintenance and network expansion.

8. Fiber Distribution Hub (FDH) Engineering

The Fiber Distribution Hub (FDH) serves as one of the most important components within the FTTH network architecture. It functions as the primary aggregation and distribution point between feeder and distribution networks.

An FDH typically contains:

- Feeder fiber terminations
- Distribution fiber terminations
- Splitter modules
- Patch panels
- Fiber management systems
- Cable routing infrastructure

The placement of FDHs significantly influences network performance, serviceability, maintenance efficiency, and future scalability.

FDH Planning Considerations

Several engineering factors were considered during FDH planning:

Service Coverage

Each FDH was positioned to efficiently serve surrounding service areas while minimizing unnecessary fiber lengths.

Capacity Requirements

FDHs were designed with sufficient capacity to accommodate both existing and projected customer demand.

Accessibility

Locations were selected to ensure ease of maintenance, future upgrades, and operational access.

Redundancy and Expansion

Spare capacity was incorporated wherever possible to support future network growth.

FDH Service Boundaries

GIS-based FDH boundaries were developed to clearly define service zones and simplify infrastructure management.

These boundaries help planners:

- Identify customer coverage areas.
- Estimate infrastructure requirements.
- Manage network capacity.
- Support future expansion planning.

The resulting FDH boundary maps provide both engineering and operational visibility into the network structure.

9. Fiber Access Terminal (FAT) Planning

Fiber Access Terminals (FATs) act as intermediate distribution points within the FTTH network. FATs bridge the connection between distribution fibers and customer access networks.

The primary functions of FATs include:

- Distribution management
- Customer aggregation
- Fiber termination
- Service provisioning

Proper FAT placement is essential because it directly influences customer connectivity, installation costs, and maintenance efficiency.

FAT Design Strategy

The FAT allocation process considered:

- Housing density
- Road accessibility
- Service demand
- Future growth potential
- Distribution cable routing

GIS analysis was used to identify optimal FAT locations while maintaining balanced service areas.

FAT Capacity Analysis

Each FAT was evaluated based on:

- Available ports
- Connected customers
- Spare capacity
- Future service demand

These metrics were later incorporated into the Power BI dashboard to support infrastructure monitoring and capacity management.

10. Network Access Point (NAP) Planning

The Network Access Point (NAP) represents the final passive distribution point before customer premises.

NAPs play a critical role in service delivery because they directly influence customer installation efficiency and network serviceability.

The NAP planning process focused on:

- Maximizing customer coverage
- Minimizing drop cable lengths
- Reducing infrastructure costs
- Improving maintenance accessibility

Serviceability Analysis

Serviceability refers to the ability of the network to provide broadband services to potential customers within a given area.

GIS-based serviceability analysis was performed to evaluate:

- Homes passed
- Potential customer coverage
- Service readiness
- Infrastructure availability
- Future expansion opportunities

Dedicated NAP boundaries were developed to support planning and operational decision-making.

The serviceability map produced as part of this project provides a clear representation of network coverage and customer accessibility throughout the study area.

11. Duct Infrastructure Planning

Duct infrastructure forms the physical pathway through which fiber optic cables are deployed.

Proper duct planning is essential because it directly affects deployment costs, installation efficiency, maintenance activities, and future network expansion.

The project included multiple duct categories:

1-Way Duct

Used in lower-density service areas where a single conduit is sufficient.

2-Way Duct

Provides additional flexibility for future expansion and maintenance activities.

4-Way Duct

Typically deployed along primary feeder routes and high-capacity network corridors.

Duct Allocation Analysis

GIS-based infrastructure analysis was used to calculate:

- Total duct lengths
- Duct distribution by category
- Feeder duct requirements
- Distribution duct requirements

These values were later incorporated into the engineering BOQ and dashboard analytics.

Efficient duct planning reduces future excavation costs and improves long-term network scalability.

12. GIS Methodology and Data Preparation

Geographic Information Systems (GIS) formed the foundation of the planning and analytical workflow used throughout this project. While telecom engineering provides the technical framework for fiber network design, GIS enables planners to visualize, analyze, manage, and optimize network infrastructure spatially.

The project datasets were initially maintained within geodatabases and engineering design files. These datasets contained information related to fiber routes, ducts, splice closures, service boundaries, network nodes, and supporting infrastructure.

Data Preparation

The first stage of the workflow involved extracting and organizing data from the geodatabase into thematic layers.

The network was categorized into the following infrastructure layers:

- Feeder Fiber Network
- Distribution Fiber Network
- Duct Infrastructure
- Splice Closures
- FDH Boundaries
- FAT Boundaries
- NAP Boundaries
- Utility Structures
- Supporting Infrastructure

Each layer was reviewed and validated to ensure consistency in geometry, attribute information, and network connectivity.

Symbology Development

To improve readability and engineering interpretation, custom symbology was developed for each network component.

Examples include:

- Feeder Fiber displayed separately from Distribution Fiber
- Different colors assigned to duct categories
- Distinct symbols for splice closures
- Individual representations for FDHs, FATs, and NAPs
- Service boundaries represented through polygon overlays

The objective was to create engineering-grade maps that could be easily interpreted by planners, field teams, and project managers.

Spatial Analysis

GIS analysis was used to support multiple planning activities including:

- Infrastructure inventory management
- Coverage analysis
- Serviceability assessment
- Boundary generation
- Capacity visualization
- Route optimization
- Infrastructure density analysis

This integration of GIS and telecom engineering improved decision-making throughout the planning process.

13. High-Level Design (HLD)

The High-Level Design (HLD) represents the strategic planning stage of the FTTH network. The purpose of the HLD is to establish the overall network architecture and define how major infrastructure components will interact within the service area.

Unlike detailed engineering drawings, the HLD focuses on the backbone structure of the network and provides a high-level representation of infrastructure deployment.

HLD Components

The HLD developed for this project includes:

- Feeder Fiber Routes
- Primary Duct Infrastructure
- FDH Locations
- Major Splice Closure Locations
- Backbone Connectivity
- Service Area Boundaries

Feeder Network Planning

The feeder network acts as the backbone of the FTTH architecture.

Key considerations during feeder planning included:

- Minimizing route length
- Maximizing service coverage
- Supporting future expansion
- Maintaining redundancy where feasible
- Reducing deployment costs

High-capacity fiber cables were allocated along major feeder routes to ensure adequate bandwidth availability for future subscriber growth.

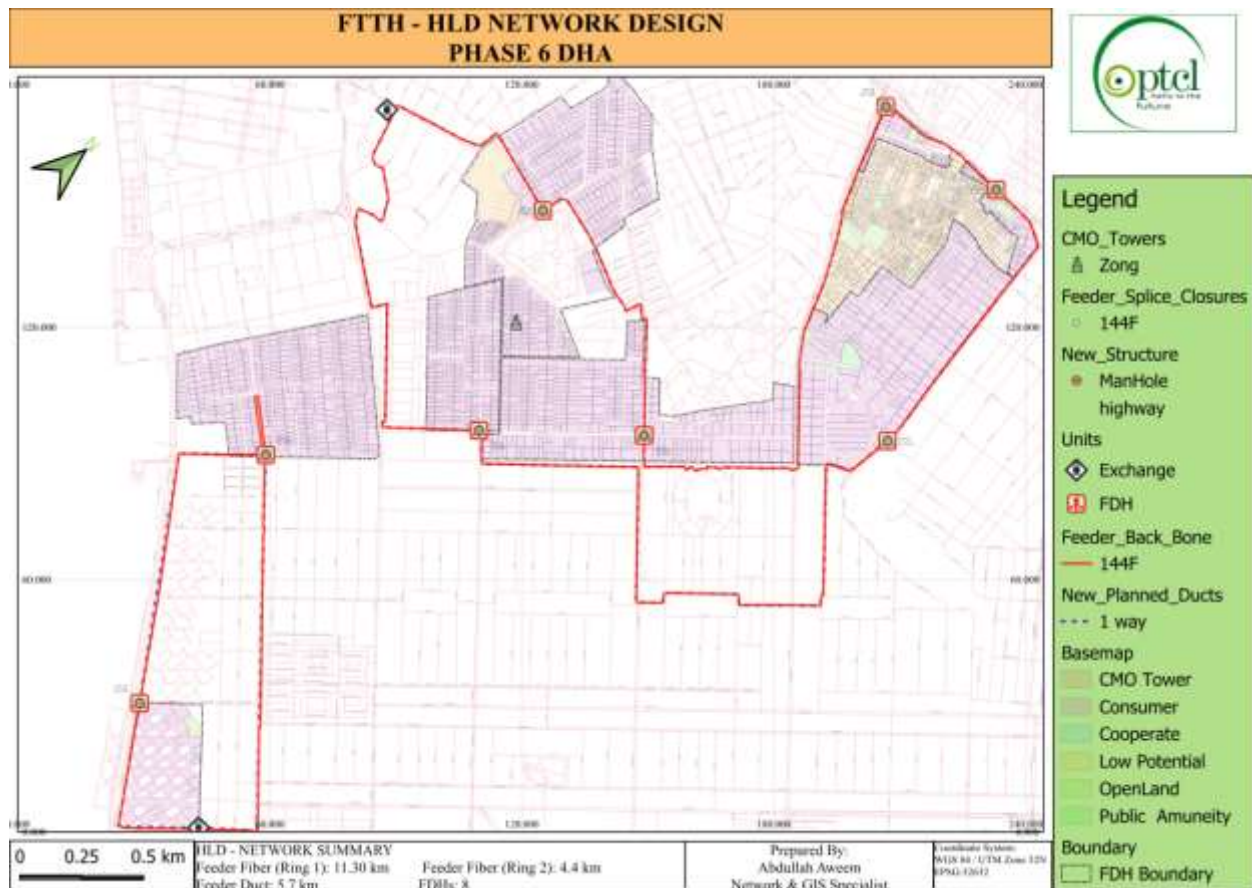
FDH Boundary Planning

Dedicated FDH service boundaries were developed using GIS analysis.

These boundaries assist with:

- Capacity planning
- Infrastructure management
- Service allocation
- Expansion planning

The HLD provides management teams with a clear overview of network structure while establishing the foundation for detailed engineering design.



HLD Outcomes

The completed High-Level Design successfully defined:

- Network backbone architecture
- Feeder routing strategy
- Service boundaries
- Primary infrastructure locations

The HLD served as the framework for subsequent Low-Level Design activities.

14. Low-Level Design (LLD)

Following completion of the High-Level Design, a detailed Low-Level Design (LLD) was developed.

The LLD represents the engineering implementation stage of the project and provides the detailed information required for field deployment.

Unlike the HLD, which focuses on strategic architecture, the LLD focuses on physical infrastructure deployment and operational implementation.

LLD Components

The Low-Level Design includes:

- Distribution Fiber Routes
- Duct Infrastructure
- FAT Locations
- NAP Locations
- Splice Closures
- Joint Locations
- Customer Service Areas
- Distribution Boundaries

Distribution Network Design

The distribution network extends connectivity from FDHs to local service areas.

Distribution planning considered:

- Customer density
- Road network geometry
- Infrastructure accessibility
- Service demand
- Future expansion potential

The distribution network was designed to maximize coverage while minimizing infrastructure requirements.

Fiber Allocation

Fiber allocation was performed according to service demand and projected growth.

The design incorporated:

- Active fibers
- Reserved fibers
- Spare fibers for maintenance
- Future growth capacity

Maintaining spare capacity is considered a best practice within telecom engineering because it reduces future upgrade costs and improves network resilience.

Splice Closure Planning

Splice closures were strategically positioned to facilitate network connectivity and maintenance operations.

Proper splice placement helps:

- Simplify network management
- Reduce optical losses
- Improve troubleshooting efficiency
- Support future expansion activities

Duct Routing

Duct routes were optimized to:

- Minimize installation costs
- Reduce excavation requirements
- Improve maintenance accessibility
- Support future infrastructure growth

The final LLD provides detailed deployment information that can be utilized directly by implementation teams.



15. Serviceability Mapping and Coverage Analysis

One of the most valuable outputs of the project is the serviceability map.

Serviceability analysis evaluates the network's ability to provide broadband services to existing and future customers within the planned service area.

Importance of Serviceability Analysis

For telecom operators, understanding service coverage is essential because it directly influences:

- Customer acquisition
- Infrastructure investment
- Capacity planning
- Expansion strategy
- Revenue generation

GIS provides a powerful platform for visualizing and evaluating serviceability.

NAP Service Areas

Dedicated NAP boundaries were developed to represent customer service zones.

These boundaries assist planners in:

- Estimating customer coverage
- Monitoring service availability
- Planning network expansion
- Evaluating infrastructure utilization

Coverage Assessment

The serviceability analysis evaluated:

- Homes Passed
- Potential Subscribers
- Infrastructure Availability
- Service Readiness
- Coverage Density

Benefits

The serviceability map enables telecom operators to:

- Identify underserved areas
- Prioritize infrastructure investments
- Improve customer acquisition strategies
- Monitor network expansion opportunities

The resulting serviceability map provides both operational and strategic value for network planning activities.



16. Bill of Quantities (BOQ) Development and Infrastructure Estimation

A Bill of Quantities (BOQ) is one of the most important deliverables produced during FTTH network planning. The BOQ provides a detailed estimate of the materials, infrastructure components, and resources required for network deployment.

Accurate BOQ preparation allows telecom operators to estimate project costs, optimize resource allocation, evaluate deployment feasibility, and support procurement activities.

The BOQ developed for this project was generated using GIS-derived measurements and engineering design outputs produced during the HLD and LLD phases.

Purpose of BOQ Development

The primary objectives of the BOQ were:

- Estimate deployment costs.
- Support procurement planning.
- Identify infrastructure requirements.
- Assist project scheduling.
- Facilitate contractor planning.
- Support future network expansion activities.

Infrastructure Components Considered

The BOQ included the following infrastructure elements:

Fiber Optic Cables

The network was divided into:

- Feeder Fiber Cables
- Distribution Fiber Cables
- Drop Fiber Cables

Cable quantities were calculated using GIS-based length measurements.

Duct Infrastructure

The design included multiple duct categories:

- 1-Way Duct
- 2-Way Duct
- 4-Way Duct

Duct quantities were calculated to support both current deployment requirements and future expansion needs.

Network Nodes

Infrastructure quantities included:

- Fiber Distribution Hubs (FDHs)
- Fiber Access Terminals (FATs)
- Network Access Points (NAPs)
- Splice Closures
- Handholes
- Joint Boxes

Passive Components

The following passive network elements were considered:

- Optical Splitters
- SC/APC Connectors
- Fusion Splice Requirements
- Drop Cable Termination Hardware

BOQ Estimation Methodology

GIS was used extensively to calculate:

- Total fiber lengths
- Duct lengths
- Infrastructure counts
- Service boundaries
- Coverage areas

These calculations reduced manual estimation errors and improved overall planning accuracy.

The final BOQ serves as a practical deployment document supporting engineering, procurement, and project management teams.

17. Power BI Dashboard Development

Modern telecom projects generate large volumes of engineering and operational data. While traditional reports provide valuable information, dashboards enable decision-makers to quickly understand network performance, infrastructure utilization, and deployment progress.

To enhance project visibility and support data-driven decision-making, a comprehensive Power BI dashboard was developed.

The dashboard integrates GIS outputs, infrastructure inventories, and engineering metrics into an interactive reporting environment.

Dashboard Objectives

The dashboard was designed to:

- Visualize telecom infrastructure assets.
- Monitor network capacity.
- Analyze fiber utilization.
- Track serviceability indicators.
- Support engineering decision-making.
- Improve management reporting.

Dashboard Structure

The dashboard was divided into three primary sections.

Executive Overview

The Executive Overview page provides a high-level summary of the network.

Key indicators include:

- Total Fiber Length
- Total Duct Length
- Number of FDHs
- Number of FATs
- Number of NAPs
- Total Connections
- Infrastructure Coverage

This page provides management teams with a quick overview of the network status.

Telecom Engineering Analytics

The engineering page focuses on technical performance metrics.

Key visualizations include:

- Core Utilization
- Fiber Capacity Analysis
- FDH Load Distribution
- FAT Utilization
- Duct Infrastructure Analysis
- Link Loss Monitoring

These metrics help engineering teams evaluate network efficiency and identify potential bottlenecks.

GIS Overview

The GIS page integrates spatial outputs into the dashboard environment.

The page includes:

- High-Level Design Map
- Low-Level Design Map
- Serviceability Map

This integration provides both technical and geographical context for decision-makers.

Benefits of Dashboard Analytics

The Power BI dashboard transforms engineering datasets into actionable intelligence by:

- Improving visibility of infrastructure assets.
- Supporting capacity planning.
- Identifying network utilization trends.
- Enhancing management reporting.
- Simplifying engineering analysis.

The dashboard serves as a centralized platform for monitoring and evaluating network performance.

18. Results and Findings

The project successfully delivered a complete FTTH planning framework integrating GIS, telecom engineering, and business intelligence technologies.

The final outputs include:

- High-Level Design (HLD)
- Low-Level Design (LLD)
- Serviceability Analysis
- Infrastructure BOQ
- Telecom Engineering Dashboard

- GIS Infrastructure Database

Network Planning Outcomes

The network design achieved the following objectives:

Optimized Infrastructure Deployment

GIS-based planning enabled efficient routing of fiber and duct infrastructure while minimizing deployment costs.

Improved Service Coverage

Serviceability analysis identified customer coverage zones and supported optimized placement of FDHs, FATs, and NAPs.

Scalable Network Architecture

The network was designed with future growth in mind by incorporating spare capacity and scalable infrastructure.

Enhanced Visibility

The integration of GIS and Power BI improved infrastructure visibility and operational awareness.

Data-Driven Decision Making

Engineering and management teams can utilize dashboard insights to support planning, maintenance, and expansion activities.

Technical Achievements

The project demonstrates practical application of:

- FTTH Network Planning
- GPON Architecture Design
- GIS-Based Telecom Engineering
- Spatial Analysis
- Fiber Infrastructure Management
- Capacity Planning
- Business Intelligence Reporting

The integration of these disciplines highlights the value of combining engineering design with geospatial technologies and analytical tools.

Overall, the project provides a scalable and replicable framework for future FTTH deployments and infrastructure planning initiatives.

19. Future Enhancements and Recommendations

Although the current project successfully delivers a complete FTTH planning framework, several opportunities exist to further enhance the network and improve operational efficiency.

The future of telecommunications is increasingly dependent on automation, real-time monitoring, geospatial intelligence, and data-driven decision-making. By integrating emerging technologies with existing GIS and telecom engineering workflows, network operators can significantly improve service quality, operational visibility, and infrastructure utilization.

WebGIS Deployment

One of the most valuable future enhancements is the development of a WebGIS platform.

Currently, engineering data is maintained within desktop GIS environments such as QGIS and ArcGIS. While these tools provide powerful analytical capabilities, they are primarily intended for GIS specialists and planners.

A WebGIS platform would enable:

- Interactive infrastructure visualization.
- Remote access to network assets.
- Field data collection and updates.
- Real-time infrastructure monitoring.
- Management reporting.
- Asset inventory management.

By deploying network data through a web-based platform, engineering teams, managers, contractors, and operations staff can access the same information through a centralized environment.

PostGIS Spatial Database Integration

The current project relies on traditional GIS datasets and engineering spreadsheets. Future deployments should migrate network assets into a centralized PostgreSQL/PostGIS database.

Benefits include:

- Centralized asset management.
- Multi-user editing capabilities.
- Improved data integrity.
- Faster querying and analysis.
- Enterprise-level scalability.
- Integration with WebGIS applications.

A PostGIS-enabled environment would provide a robust spatial database foundation capable of supporting future telecom operations.

Real-Time Asset Management

Future network operations can benefit from the implementation of a centralized asset management system.

Such a system could manage:

- Fiber routes.
- FDHs.
- FATs.
- NAPs.
- Splice closures.
- Customer connections.
- Maintenance records.

Integration with GIS would allow engineering teams to quickly identify affected infrastructure during outages, maintenance activities, or expansion projects.

Network Digital Twin

A Digital Twin represents a virtual representation of physical infrastructure.

The development of a Digital Twin for the FTTH network would allow operators to:

- Simulate network expansion.
- Model service demand growth.
- Analyze infrastructure utilization.
- Predict congestion.
- Evaluate redundancy scenarios.

Telecom operators to improve planning and operational decision-making are increasingly adopting digital Twin technology.

Predictive Analytics and Machine Learning

Future versions of the project can incorporate predictive analytics using machine-learning models.

Potential applications include:

- Customer demand forecasting.
- Capacity planning.
- Infrastructure utilization prediction.
- Fault prediction.
- Maintenance scheduling.

These capabilities would further enhance the strategic value of the network-planning framework.

Mobile GIS Integration

Field survey and maintenance activities can be significantly improved through mobile GIS solutions.

Future implementations may include:

- Survey123
- Field Maps
- Mobile GIS applications
- GPS-enabled inspections

These tools would streamline field operations and improve data accuracy.

20. Conclusion

The rapid growth of digital services has transformed broadband connectivity from a convenience into a critical infrastructure requirement. Fiber to the Home (FTTH) technology has emerged as the most effective solution for meeting these demands due to its high bandwidth capacity, scalability, reliability, and long-term sustainability.

This project successfully demonstrates the integration of Geographic Information Systems (GIS), GPON engineering principles, infrastructure planning methodologies, and business intelligence analytics within a unified FTTH planning framework.

The project began with field data collection and infrastructure assessment, followed by network architecture development, feeder and distribution planning, optical network design, serviceability analysis, and infrastructure estimation. GIS technologies were used extensively throughout the planning process to support spatial analysis, network visualization, asset management, and engineering decision-making.

The resulting deliverables included:

- High-Level Design (HLD)
- Low-Level Design (LLD)
- Serviceability Analysis
- Fiber Infrastructure BOQ
- Telecom Engineering Dashboard
- GIS-Based Asset Framework

The Power BI dashboard further enhanced the project by transforming engineering datasets into actionable insights for both technical and management stakeholders.

From an engineering perspective, the project demonstrates the practical application of GPON network design concepts including feeder planning, splitter allocation, FDH dimensioning, FAT and NAP deployment, optical power budgeting, and capacity management.

From a GIS perspective, the project highlights the importance of spatial analysis in infrastructure planning, service coverage evaluation, and network optimization.

The combination of telecom engineering, GIS analysis, and business intelligence reporting creates a scalable framework capable of supporting future broadband deployments and network expansion initiatives.

Beyond its technical outputs, this project reflects my professional experience in GIS, telecom engineering, infrastructure analytics, and data-driven decision-making. It demonstrates my ability to combine engineering principles with modern geospatial technologies to develop practical solutions for real-world infrastructure challenges.

As broadband networks continue to evolve, the integration of GIS, WebGIS, spatial databases, analytics, and automation will play an increasingly important role in supporting smarter and more efficient telecommunications infrastructure.

This project serves as a foundation for future developments involving enterprise GIS platforms, PostGIS databases, WebGIS applications, digital twins, and advanced network analytics.

21. References

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